

Enumerate first; audit only with the residual physical-call budget

1. Validate

Replay empty + full
invalid: out of scope

2. LOO test

Delete each document
from backbone B

3. Cause search

Protect enumeration
exhaust its frontier

4. Safety audit

Sample unseen classes
stop at a reversal

5. Grade output

Causes + witness + cost
certify or abstain

A found reversal rejects monotonicity; an unviolated non-exhaustive trace remains inconclusive.